

A PROOF OF THE CHAUHAN–SHUKLA–VINAYAK CONJECTURE ON THE 3-CUT COMPLEX OF $K_m \square P_n$

ABSTRACT. For a graph G and $k \geq 1$, the k -cut complex $\Delta_k(G)$ is the simplicial complex whose facets are the sets $\sigma \subseteq V(G)$ with $|\sigma| = |V(G)| - k$ for which $G[V(G) \setminus \sigma]$ is disconnected. Chauhan, Shukla and Vinayak conjectured a closed form for $\tilde{H}_*(\Delta_3(K_m \square P_n))$ for all $m, n \geq 3$. We prove it. The combinatorial Alexander dual of $\Delta_3(K_m \square P_n)$ is the complex whose *maximal* faces of dimension at least two are exactly n column simplices, $(n-1)\binom{m}{2}$ squares and $m(n-2)$ row triangles, sitting over the complete 1-skeleton, and that complex has the homotopy type of a wedge of $(n-1)\binom{m-1}{2}$ two-spheres and $\frac{m}{2}(mn-m-2)(n-2)$ circles. All groups are free; there is no torsion in any degree.

1. THE CONJECTURE

Let G be a graph on N vertices and let $1 \leq k \leq N$. Following Bayer, Denker, Jelić Milutinović, Rowlands, Sundaram and Xue [1], the *k -cut complex* $\Delta_k(G)$ is the simplicial complex on $V(G)$ whose facets are the sets $\sigma \subseteq V(G)$ with $|\sigma| = N - k$ such that the induced subgraph $G[V(G) \setminus \sigma]$ is disconnected. Equivalently, the facets are the complements of the k -subsets that induce a disconnected subgraph, so $\Delta_k(G)$ is pure of dimension $N - k - 1$ whenever it is non-empty.

Write $K_m \square P_n$ for the Cartesian product of the complete graph K_m with the path on n vertices: the vertex set is

$$(1) \quad V = \{0, \dots, m-1\} \times \{0, \dots, n-1\},$$

and, for distinct vertices,

$$(2) \quad (i, j) \sim (i', j') \iff \text{either } j = j' \text{ and } i \neq i', \text{ or } i = i' \text{ and } |j - j'| = 1.$$

So $K_m \square P_n$ is n disjoint copies of K_m , which we call *columns*, joined by perfect matchings between consecutive columns. Throughout, $N = mn$, $G = K_m \square P_n$ and $\Delta_3 = \Delta_3(G)$, which is pure of dimension $mn - 4$.

Chauhan, Shukla and Vinayak [4, Conjecture 4.4] state the following, on the basis of SageMath computations of $\tilde{H}_*(\Delta_3(K_m \square P_n))$ for small m and n .

2020 *Mathematics Subject Classification.* 05E45, 55U10, 05C76.

Key words and phrases. cut complex, Alexander dual, simplicial homology, Cartesian product of graphs, Betti number, nerve lemma.

Conjecture 4.4 of Chauhan, Shukla and Vinayak. For $m, n \geq 3$,

$$\tilde{H}_i(\Delta_3(K_m \square P_n)) = \begin{cases} \mathbb{Z}^{\frac{1}{2}(m^2-3m+2)(n-1)} & \text{if } i = mn - 5, \\ \mathbb{Z}^{\frac{m}{2}(mn-m-2)(n-2)} & \text{if } i = mn - 4, \\ 0 & \text{otherwise.} \end{cases}$$

Put

$$(3) \quad \begin{aligned} A(m, n) &= \frac{1}{2}(m^2 - 3m + 2)(n - 1) = (n - 1) \binom{m - 1}{2}, \\ B(m, n) &= \frac{m}{2}(mn - m - 2)(n - 2). \end{aligned}$$

Theorem 1. For all $m, n \geq 3$ and with integer coefficients,

$$\tilde{H}_i(\Delta_3(K_m \square P_n)) = \begin{cases} \mathbb{Z}^{A(m, n)} & i = mn - 5, \\ \mathbb{Z}^{B(m, n)} & i = mn - 4, \\ 0 & \text{otherwise,} \end{cases}$$

and both groups are free: there is no torsion in any degree. In particular Conjecture 4.4 of [4] holds.

2. THE ALEXANDER DUAL

For a complex Δ on a vertex set V with $|V| = N$, the *combinatorial Alexander dual* is $\Delta^\vee = \{S \subseteq V : V \setminus S \notin \Delta\}$, and Alexander duality [5, Theorem 5.6] gives, for every i ,

$$(4) \quad \tilde{H}_i(\Delta; \mathbb{Z}) \cong \tilde{H}^{N-i-3}(\Delta^\vee; \mathbb{Z}).$$

Write $\Delta_3^\vee$ for the dual of $\Delta_3(G)$ on V .

Lemma 2. $S \in \Delta_3^\vee$ if and only if every 3-subset of S induces a connected subgraph of G , equivalently if and only if the non-edges of $G[S]$ form a matching.

Proof. $V \setminus S \in \Delta_3$ means that $V \setminus S$ lies in some facet $V \setminus T$ with $|T| = 3$ and $G[T]$ disconnected, which happens if and only if $T \subseteq S$ for some such T . Hence $S \in \Delta_3^\vee$ if and only if no 3-subset of S induces a disconnected subgraph. A graph on three vertices is connected exactly when it has at least two edges, i.e. at most one non-edge; so the condition says that no two non-edges of $G[S]$ lie inside a common triple, i.e. that no two non-edges of $G[S]$ share a vertex. $\square$

Equivalently: *every vertex of S has at most one non-neighbour in S* . We use that form. Write $\Sigma_j = \{0, \dots, m - 1\} \times \{j\}$ for the j -th column.

Lemma 3. Let $m, n \geq 3$. Then $\Delta_3^\vee$ contains all $\binom{mn}{2}$ pairs of vertices, and its maximal faces of dimension at least 2 are exactly the following three families:

- the n column simplices Σ_j , of dimension $m - 1$ ($G[\Sigma_j] = K_m$, no non-edge);

- the $(n-1)\binom{m}{2}$ squares $Q_{j,\{a,b\}} = \{a,b\} \times \{j, j+1\}$, of dimension 3, for $0 \leq j \leq n-2$ and $a < b$ (two disjoint non-edges, $(a,j)(b,j+1)$ and $(b,j)(a,j+1)$);
- the $m(n-2)$ row triangles $T_{i,j} = \{(i,j), (i,j+1), (i,j+2)\}$, of dimension 2, for $0 \leq j \leq n-3$ (one non-edge, $(i,j)(i,j+2)$).

Every other maximal face of $\Delta_3^\vee$ is an edge.

Proof. Each pair of vertices spans at most one non-edge, so all pairs are faces by Lemma 2, and each of the three listed sets is a face by the parenthetical computation beside it. Conversely let $S \in \Delta_3^\vee$ with $|S| \geq 3$ and let $J = \{j : S \cap \Sigma_j \neq \emptyset\}$. Note from (2) that for $j < j'$ the vertices (i,j) and (i',j') are adjacent if and only if $i = i'$ and $j' = j+1$.

If $|J| = 1$ then $S \subseteq \Sigma_j$.

Let $|J| = 2$, say $J = \{j, j'\}$ with $j < j'$, and put $p = |S \cap \Sigma_j|$, $q = |S \cap \Sigma_{j'}|$. If $j' \geq j+2$ then every one of the pq cross pairs is a non-edge, so each of the $p+q \geq 3$ vertices has $\geq \min(p, q) \geq 1$ non-neighbours and some vertex has ≥ 2 ; impossible. So $j' = j+1$. A vertex $(i, j) \in S$ is non-adjacent to every $(i', j+1) \in S$ with $i' \neq i$, hence has at least $q-1$ non-neighbours, so $q \leq 2$; symmetrically $p \leq 2$. If $p = q = 2$, with rows $\{a, b\}$ in column j and $\{c, d\}$ in column $j+1$, then $a \in \{c, d\}$ and $b \in \{c, d\}$ (else (a, j) or (b, j) has two non-neighbours), so $\{c, d\} = \{a, b\}$ and $S = Q_{j, \{a, b\}}$. If $p = 2, q = 1$, with rows $\{a, b\}$ and $\{c\}$, then $c \in \{a, b\}$ and $S \subseteq Q_{j, \{a, b\}}$; the case $p = 1, q = 2$ is symmetric.

Let $|J| \geq 3$ and let $j_1 < j_2 < j_3$ be the three smallest elements of J . Fix $u \in S \cap \Sigma_{j_1}$. Every vertex of $S \cap \Sigma_{j_3}$ is a non-neighbour of u , since $j_3 \geq j_1+2$; and every vertex of $S \cap \Sigma_{j_2}$ is a non-neighbour of u except, when $j_2 = j_1+1$, the one in u 's own row. As u has at most one non-neighbour and both $S \cap \Sigma_{j_2}$ and $S \cap \Sigma_{j_3}$ are non-empty, this forces $j_2 = j_1+1$, $|S \cap \Sigma_{j_2}| = 1$ with that vertex in u 's row, $|S \cap \Sigma_{j_3}| = 1$, and $|J| = 3$ (a fourth column $j_4 \geq j_1+3$ would give u a second non-neighbour). Running the same argument downwards from $w \in S \cap \Sigma_{j_3}$ gives $j_2 = j_3-1$, $|S \cap \Sigma_{j_1}| = 1$ and the Σ_{j_2} vertex in w 's row. Hence $j_3 = j_1+2$ and S is one vertex in each of three consecutive columns, all in one row: $S = T_{i, j_1}$. $\square$

3. PROOF OF THEOREM 1

Fix $m, n \geq 3$ and let $Y \subseteq \Delta_3^\vee$ be the subcomplex generated by the faces of dimension at least 2, that is, by the three families of Lemma 3. We use twice the following standard fact: if a simplicial complex is $X = X_1 \cup X_2$ with $X_1 \cap X_2$ a non-empty contractible subcomplex, then $X \simeq X_1 \vee X_2$; in particular if X_2 is contractible then $X \simeq X_1$. (Collapsing the contractible subcomplex $X_1 \cap X_2$ is a homotopy equivalence, and the quotient is the wedge.)

Step 1: the edges, and $\Delta_3^\vee \simeq Y \vee \bigvee_E S^1$. An edge of Y lies in one of the three generating families, and conversely, so the edges of Y are exactly

$$(5) \quad \underbrace{n \binom{m}{2}}_{\text{inside a column}} + \underbrace{m^2(n-1)}_{\text{between consecutive columns}} + \underbrace{m(n-2)}_{\text{skip edges } (i,j)(i,j+2)}.$$

Indeed Σ_j contributes precisely the column pairs; $Q_{j,\{a,b\}}$ contributes column pairs together with the four cross pairs $\{a,b\} \times \{j\} \rightarrow \{a,b\} \times \{j+1\}$, and as $\{a,b\}$ runs over all pairs these give every one of the m^2 pairs between columns j and $j+1$ (for $a=b$ use any $Q_{j,\{a,c\}}$, $c \neq a$, which needs $m \geq 2$); and $T_{i,j}$ contributes two consecutive-column pairs and the skip pair $(i,j)(i,j+2)$. No generator contains a pair of vertices in columns at distance ≥ 3 , or at distance 2 in different rows. The three classes in (5) are disjoint, so

$$(6) \quad E := \binom{mn}{2} - n \binom{m}{2} - m^2(n-1) - m(n-2)$$

pairs of vertices are edges of $\Delta_3^\vee$ but not of Y . Now Y contains every vertex and is connected (each column is a simplex and consecutive columns are joined), and by Lemma 3 $\Delta_3^\vee$ is Y with those E further 1-cells attached. Attaching a 1-cell to a connected complex along two of its vertices is, up to homotopy, wedging on a circle; hence

$$\Delta_3^\vee \simeq Y \vee \bigvee_E S^1.$$

Step 2: $E = B(m,n)$, identically. Both sides of (6) and (3) are polynomials in m and n , and both expand to

$$\frac{1}{2}(m^2n^2 - 3m^2n + 2m^2 - 2mn + 4m),$$

so $E = B(m,n)$ for all m,n ; this is an identity of coefficients, not a check at sample values.

Step 3: the two-column block $\mathcal{B}_k \simeq \bigvee_{\binom{m-1}{2}} S^2$. For $0 \leq k \leq n-2$ let $\mathcal{B}_k \subseteq Y$ be the subcomplex generated by Σ_k , Σ_{k+1} and the $\binom{m}{2}$ squares $Q_{k,\{a,b\}}$.

Cover $\mathcal{B}_k$ by those $\binom{m}{2} + 2$ simplices. Every non-empty intersection of a subfamily is again a simplex, hence contractible: the intersection of simplices is the simplex on the intersection of their vertex sets, and here $\Sigma_k \cap \Sigma_{k+1} = \emptyset$, $\Sigma_k \cap Q_{k,\{a,b\}}$ is the edge $\{a,b\} \times \{k\}$, $Q_{k,\{a,b\}} \cap Q_{k,\{c,d\}}$ is the simplex on $(\{a,b\} \cap \{c,d\}) \times \{k, k+1\}$, and adjoining Σ_k or Σ_{k+1} to a subfamily of squares with common row set $R = \bigcap \{a_t, b_t\}$ gives $R \times \{k\}$ or $R \times \{k+1\}$. By the nerve lemma [3, Theorem 10.6], $\mathcal{B}_k$ is homotopy equivalent to the nerve $\mathcal{N}$ of this cover.

Let $\mathcal{K}$ be the simplicial complex whose vertices are the $\binom{m}{2}$ pairs $\{a,b\} \subseteq [m]$ and whose faces are the families of pairs having a common element. Reading off the previous paragraph: a family of squares is a face of $\mathcal{N}$ if and only if its row sets have a common element, i.e. $\mathcal{K}$ is the induced subcomplex

of $\mathcal{N}$ on the square-vertices; each such face may be enlarged by Σ_k or by Σ_{k+1} but not by both, and $\{\Sigma_k, \Sigma_{k+1}\}$ alone is not a face. Hence $\mathcal{N}$ is the union of two cones over $\mathcal{K}$ glued along $\mathcal{K}$, i.e. $\mathcal{N} = \text{Susp}(\mathcal{K})$.

Finally, apply the nerve lemma a second time, to $\mathcal{K}$. For $a \in [m]$ let Λ_a be the full simplex spanned by the $m-1$ pairs containing a . Each face of $\mathcal{K}$ has a common element a and so is a face of Λ_a ; hence the Λ_a cover $\mathcal{K}$. Again the intersection of full simplices is the simplex on the intersection of their vertex sets, so for distinct a, b, c we get $\Lambda_a \cap \Lambda_b = \{\{a, b\}\}$, a single vertex, and $\Lambda_a \cap \Lambda_b \cap \Lambda_c = \emptyset$. Every non-empty intersection of a subfamily is therefore a simplex, and the nerve of this cover has one vertex for each $a \in [m]$, one edge for each pair $\{a, b\}$ and no face of dimension ≥ 2 : it is the graph K_m . So $\mathcal{K} \simeq K_m$, a connected graph with m vertices and $\binom{m}{2}$ edges, whence $\mathcal{K} \simeq \bigvee_{\binom{m}{2}-m+1} S^1 = \bigvee_{\binom{m-1}{2}} S^1$ and

$$(7) \quad \mathcal{B}_k \simeq \text{Susp}\left(\bigvee_{\binom{m-1}{2}} S^1\right) \simeq \bigvee_{\binom{m-1}{2}} S^2.$$

In particular $\tilde{H}_1(\mathcal{B}_k) = 0$, $\tilde{H}_2(\mathcal{B}_k) = \mathbb{Z}^{\binom{m-1}{2}}$ and $\tilde{H}_j(\mathcal{B}_k) = 0$ otherwise, all free.

Step 4: $Y \simeq \bigvee_A S^2$. Let $Y_0 = \mathcal{B}_0 \cup \dots \cup \mathcal{B}_{n-2}$, which is the subcomplex of Y generated by the columns and the squares. Set $X_0 = \mathcal{B}_0$ and $X_{t+1} = X_t \cup \mathcal{B}_{t+1}$. The vertices of X_t lie in columns $0, \dots, t+1$ and those of $\mathcal{B}_{t+1}$ in columns $t+1, t+2$, so $X_t \cap \mathcal{B}_{t+1}$ is a subcomplex of the simplex Σ_{t+1} , and it equals Σ_{t+1} because Σ_{t+1} lies in both. That is non-empty and contractible, so $X_{t+1} \simeq X_t \vee \mathcal{B}_{t+1}$ and, by (7) and induction,

$$Y_0 \simeq \bigvee_{k=0}^{n-2} \mathcal{B}_k \simeq \bigvee_{(n-1)\binom{m-1}{2}} S^2 = \bigvee_{A(m,n)} S^2.$$

Now adjoin the $m(n-2)$ row triangles one at a time, in any order. When $T_{i,j}$ is adjoined, its edges $(i, j)(i, j+1)$ and $(i, j+1)(i, j+2)$ are already present — they are consecutive-column edges, from $\mathcal{B}_j$ and $\mathcal{B}_{j+1}$ — while its third edge $(i, j)(i, j+2)$ is not, by (5) and because distinct row triangles have distinct skip edges. So the intersection of the (contractible) 2-simplex $T_{i,j}$ with the complex built so far is the path $(i, j) - (i, j+1) - (i, j+2)$, which is contractible, and the homotopy type is unchanged. Hence

$$Y \simeq Y_0 \simeq \bigvee_{A(m,n)} S^2.$$

Step 5: Alexander duality. Combining Steps 1, 2 and 4,

$$\Delta_3^\vee \simeq \left(\bigvee_{A(m,n)} S^2\right) \vee \left(\bigvee_{B(m,n)} S^1\right),$$

so $\tilde{H}_1(\Delta_3^\vee) = \mathbb{Z}^B$, $\tilde{H}_2(\Delta_3^\vee) = \mathbb{Z}^A$ and all other reduced homology vanishes; every group is free, so by the universal coefficient theorem the reduced cohomology is $\tilde{H}^1 = \mathbb{Z}^B$, $\tilde{H}^2 = \mathbb{Z}^A$ and 0 elsewhere, with no Ext contribution. Feeding this into (4) with $N = mn$: the degree $j = 1$ gives $i = mn - 4$ and the group $\mathbb{Z}^{B(m,n)}$, the degree $j = 2$ gives $i = mn - 5$ and the group

$\mathbb{Z}^{A(m,n)}$, and every other degree gives 0. This is exactly the statement of Theorem 1. $\square$

4. SCOPE

Theorem 1 settles Conjecture 4.4 of [4] as stated, for every $m, n \geq 3$, with integer coefficients and with both groups free. Three limits should be stated.

- The boundary lines $n = 2$ and $m = 2$ lie outside $m, n \geq 3$ and are separately in print [1, 2]; the closed form (3) agrees with them, and we claim no priority for those two lines.
- Only the *ordinary* 3-cut complex is treated. The total k -cut complex is a different object, and the sibling Conjectures 4.3, 4.5 and 4.6 of [4] are untouched here.
- Nothing here says $K_m \square P_n$ is the largest family to which the argument applies. The proof uses only Lemma 3, so it will transfer to any graph whose 3-cut dual decomposes into finitely many shapes; we have not looked for the general statement.

Artifact. The proof above is by hand and needs no computation. Accompanying this note are `verify.py` (Python 3.9+, standard library only, no external data file, exact integer arithmetic, homology by an integer Smith normal form so that torsion is detected rather than assumed away) and the transcript `verify.output.txt` of a recorded run of it. The program checks ingredients of the proof at finitely many pairs (m, n) : the identification of $\Delta_3^\vee$ in Lemma 3, the polynomial identities of (3) and (6) and the duality degree shift, and the reduced homology of $\mathcal{B}_k$, of Y and of $\Delta_3^\vee$, together with Δ_3 itself computed directly at the three smallest cells. No finite computation can verify a statement about an infinite family; the theorem is settled by the proof above, and the accompanying `REVIEW_NOTE.md` records what the program does not cover.

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